Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25/10/2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

1. Team & Project Overview

Team Name	Code Queens
Project / Solution Name	NextArc
Name (Leader)	Chew Khai Shuen
Email & Phone (Leader)	23034093@imail.sunway.edu.my 012-4085129
University Affiliation(s)	Sunway University
Live Prototype URL	https://sdc-hackathon-code-queens-1.replit.app/
Code Repository URL (public / access granted to judges)	https://github.com/khaishuen819/SDC_Hackathon_CodeQueens

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

NextArc serves a plain HTML/CSS/JavaScript interface from a Node.js/Express/TypeScript backend on Replit. The preliminary pilot supports CS fresh graduates exploring Backend Development and QA Automation. Fictional PDF/TXT CVs are processed in memory. With Gemini extraction enabled, CV text and catalogue context are sent to Google for a structured draft profile; users correct and confirm their skills before recommendations.

The backend computes explainable one-point confirmed-skill coverage and ordered learning pathways while preserving the selected route. Prisma connects to Neon PostgreSQL, which stores public reference content, immutable criteria snapshots and refresh-control state, not CVs or personal profiles. Results, pathways, citations and SDG explanations use the same criteria version.

An implemented update pipeline retrieves GitHub Changelog and JetBrains AI feeds, optionally interprets them using Gemini, validates proposed capability changes and publishes coherent snapshots. Automatic trend interpretation is currently disabled in production; the deployed app serves a reviewed baseline recognising 16 capabilities. GitHub stores source code, dependency locks and tests.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
HTML, CSS and JavaScript ES modules	Team-authored browser interface and interaction logic.	Apache-2.0 (team code)
Node.js 24.x	Backend runtime; built-in test runner.	MIT; bundled notices
TypeScript 7.0.2	Backend types and static checking.	Apache-2.0

Name & Version	Purpose in Project	License
Express 5.3.0	HTTP API, uploads route and static frontend serving.	MIT
Zod 4.6.7	Profile, request and model-output validation.	MIT
Multer 2.4.0	Multipart CV upload held in memory.	MIT
unpdf 1.9.0	Extract text from supported PDF uploads.	MIT
PDF.js 6.4.299 (bundled by unpdf)	PDF parsing used through unpdf.	Apache-2.0
@google/genai 2.30.0	Google Gemini SDK for CV-to-profile extraction and the configurable public-source interpretation adapter. Production trend interpretation is currently disabled.	Apache-2.0; SDK licence is separate from Gemini API service terms.
Prisma 7.10.1 CLI, client and pg adapter	Database schema, migrations and access to reference content, immutable criteria snapshots and refresh-control state.	Apache-2.0
pg 8.23.1	PostgreSQL database connection driver.	MIT
PostgreSQL Server version: 18.6 (hosted on Neon)	Reference database hosted on Neon.	PostgreSQL License
@types/express 5.0.6 @types/node 26.6.5 @types/multer 2.3.0 @types/pg 8.23.1	Type declarations used during development and checking.	MIT
IBM Plex Sans, Sans Condensed and Mono Google Fonts delivery	Frontend typography. CDN assets are not version-pinned.	SIL Open Font License 1.1

Name & Version	Purpose in Project	License
fast-xml-parser 5.11.2	Parses public RSS feeds for the trend pipeline, alongside application checks for size, nesting and prohibited XML declarations.	MIT
PGlite / @electric-sql/pglite — development only; package version not recorded	Temporary in-memory PostgreSQL verification during development. Not the production database and not evidence of multi-connection Neon behaviour.	Apache-2.0; bundled PostgreSQL retains its own licence.

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Gemini API / Google	Produces draft skill profiles from fictional CV text. A separate configurable adapter interprets retrieved public AI-development content; production trend interpretation is currently disabled. Mock extraction makes no model call..	Free tier, as recorded from the team's Google AI Studio account.
GitHub Changelog RSS / GitHub — https://github.blog/changelog/feed/	Configured public source for AI-tool and software-development updates. Used in development retrieval checks; live production trend interpretation is disabled.	Public feed; no API key.
JetBrains AI RSS / JetBrains — https://blog.jetbrains.com/ai/feed/	Configured public source for AI-assisted development updates. Included in the implemented retrieval pipeline; live production trend interpretation is disabled.	Public feed; no API key.
O*NET Resource Center public JSON downloads / USDOL–ETA	Development-time import of occupational task and software-reference data for the two pilot routes. This reference does not itself establish current AI trends or graduate hiring demand.	Public downloads; no API key.
Google Fonts CSS API and font CDN / Google	Loads IBM Plex font CSS and font files in the browser.	Public font delivery; no API key.

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	**Purpose in Project**	**Plan / Region**
Replit	Development workspace and Autoscale deployment of the Node/Express app and frontend.	Plan: Replit Core; Autoscale deployment (2 vCPU / 4 GiB RAM, up to 3 machines) Deployment region: Asia
Neon / PostgreSQL	Managed PostgreSQL database for public reference content, reviewed criteria snapshots, source/evidence metadata and refresh-control state. Application CV files and personal profiles are not stored in this database.	Plan: Free Database region: AWS Asia Pacific 1 (Singapore)
GitHub	Source repository, collaboration, dependency lock and test source.	Public repository. Account plan: GitHub Free Hosting region: provider-managed.

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Google Gemini Model: gemini-3.5-flash-lite CV extraction mode: gemini Production trend interpretation: disabled	Google Gemini API / Google AI Studio	Pre-trained hosted model for producing draft profiles from fictional CVs. A separate public-source interpretation adapter is implemented but disabled in production. No custom model training or fine-tuning.	Google APIs Terms of Service and Gemini API Additional Terms. Gemini API terms

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Fictional Backend/QA graduate CVs	Team repository: docs/sample-cvs/ and frontend samples. Fictional CVs	Synthetic; development, extraction examples and demonstration.	Team-authored material: Apache-2.0.	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Synthetic profiles, source/model fixtures and test scenarios	Team repository: backend tests, trend fixtures and frontend checks. Backend tests Frontend checks	Synthetic; matching, pathways, interpretation, version consistency and interface tests. Not real publisher evidence.	Team-authored material: Apache-2.0.	N
O*NET 31.0 occupational reference	O*NET Resource Center: task statements and software-skills tables. O*NET database	Public; filtered to occupations 15-1252.00 and 15-1253.00. Used as an occupational foundation, not graduate hiring-demand evidence.	CC BY 4.0; retain USDOL/ETA attribution and adaptation notice. Attribution guidance	N
Reviewed pilot capability catalogue, prerequisites, learning selections and SDG rationales	Team repository: pilot-baseline.json and baseline-provenance.json. Reviewed baseline Provenance	Team-authored/adapted reference content; 16 recognised capabilities and eight assessed per pilot route. Not measured market demand.	Team-authored portions: Apache-2.0. O*NET-derived material: CC BY 4.0. Linked content retains publisher terms.	N
Legacy placeholder seed catalogue	Team repository: prisma/seed.ts. Legacy seed	Synthetic development/demo seed content, labelled as placeholder and separate from the reviewed versioned baseline.	Team-authored material: Apache-2.0.	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
UN SDG names, numbers and Target 4.4 reference	United Nations Goal 4 reference. UN Goal 4	Public reference labels and targets. Team mappings describe possible contributions, not achieved impact.	Applicable UN reuse/copyright terms; attribution retained. Team explanations are separately authored. UN terms	N
GitHub Changelog feed content	GitHub Changelog RSS feed. GitHub feed	Public; development retrieval checks and configured trend source. Production interpretation is disabled.	Publisher copyright and applicable terms. Public access is not a blanket open licence.	N
JetBrains AI feed content	JetBrains AI RSS feed. JetBrains feed	Public; configured trend source. Production interpretation is disabled.	Publisher copyright and applicable terms; retain citations and limited supporting material.	N
Linked learning-resource references	Provider URLs recorded in baseline and provenance files. Resource provenance	Public documentation/tutorial links with team-authored practice suggestions and time estimates. No full courses are hosted.	Each provider retains its content rights and terms.	N
Aggregate manual-search pilot results	Team pilot reported in the preliminary proposal.	Self-collected evaluation: 10 participants; approximately 10-30 minutes searching; five selected a skill they considered relevant. Not model-training data.	Team-generated aggregate findings.	N for the aggregate figures submitted; no identities or CVs included.

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude Code 2.1.309 / Anthropic Model: Claude Opus 5.5.	Backend implementation, automated tests and technical documentation.	Major
Replit Agent Automatically routed model; version not recorded.	Initial server scaffold, workspace configuration and deployment assistance.	Minor
Codex / OpenAI Exact model versions not recorded.	Requirements, documentation and Feature 2 coding, testing, review and backend/frontend integration.	Major
Claude chat / Anthropic Model/version not recorded.	Problem research, requirements, deployment troubleshooting and documentation drafts.	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

We checked AI-assisted changes against our approved scope using TypeScript checks, contract validation, automated backend/frontend tests and clearly labelled fictional scenarios. Source mappings and learning links were inspected, and review findings were corrected. Users verify extracted skills before matching. The team remains responsible for understanding, reviewing and correcting generated code and documentation. Test fixtures do not establish live model quality or measured career accuracy. We distinguish implemented code, observed deployment behaviour and proposed features; live trend interpretation is currently disabled.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit, through SDC Hackathon	SDC-provided Replit Core access; stated value US$20 per participant.	Development workspace, Replit Agent assistance and Autoscale hosting of the prototype.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

SDC-provided Replit access supported our development workspace, Replit Agent assistance and Autoscale hosting of the prototype. Secrets are managed through Replit Secrets. Replit is important to this deployment, but the architecture is technically portable to another compatible Node.js host and PostgreSQL service. Gemini is accessed through an adapter rather than a self-hosted model. Technical replaceability does not remove the competition's Replit-use requirement.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Only fictional CVs are used. Uploads are processed in server memory and are not saved to application files/database or intentionally logged. With Gemini extraction enabled, CV text and catalogue context are sent to Google; input contact details are not automatically redacted. Users review the draft. Profiles, possession answers and learning choices remain in browser memory. PostgreSQL retains public references, evidence, criteria snapshots and update-control state, with no automatic snapshot expiry and no personal profiles. Enabled trend interpretation sends public source/reference material, not CVs or profiles. Secrets stay in server environment variables. Google Fonts receives browser requests. Providers may retain operational metadata under their terms. Gemini unpaid-service terms allow submitted content/responses to support service improvement; provider retention is not controlled by our code. Pilot findings are reported as aggregate counts and timings.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Gemini may misread CVs, misinterpret articles, produce unsupported mappings or fail through timeouts, quotas and provider outages. Controls include fictional-data-only use, a 2 MB upload limit, text-PDF/TXT checks, model timeouts, schema/source-excerpt validation and editable profile confirmation. Scanned PDFs have no OCR support. Trend controls include bounded retrieval, capability-deduplication and scope checks, resource/SDG readiness gates, shared refresh controls and bounded retries. Validated snapshots preserve the last accepted criteria when updates fail; storage outages can still prevent service. These safeguards do not eliminate hallucination, bias or prompt injection. Scores measure confirmed coverage of a limited two-route catalogue, not proficiency, employability or hiring probability. Useful older evidence is retained with honest dates. Automatic trend interpretation is currently disabled in production; its live recommendation quality remains unverified. Multiple future scenarios are planned for the final round.

7. Compliance & Readiness Checklist

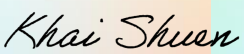
Tick all boxes that apply before submitting.

- ☒ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☒ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☒ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☒ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☒ Any use of official SDC sponsor resources has been declared in Section 5.
- ☒ The prototype described here is accessible to judges through the links provided in Section 1.
- ☒ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Signature:



Chew Khai Shuen

Date: 25/10/2026